

Correlation between Line-of-Sight Density and SNR in Binary Black Hole Mergers

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1 Introduction

This study investigates the possible correlation between astrophysical parameters of gravitational-wave events and the density of matter along the line of sight (LOS). We analyze six binary black hole mergers from LIGO/Virgo catalogs, testing whether LOS galaxy density influences the observed signal-to-noise ratio (SNR).

2 Data

- **Gravitational-wave events:** Posterior samples from the LIGO/Virgo GWOSC public releases (GWTC-1 and GWTC-2).
- **Galaxy catalog:** Sloan Digital Sky Survey (SDSS DR16).
- **Events analyzed:** GW150914, GW151012, GW170104, GW170823, GW191103, GW191109.

3 Variable Construction

- **Chirp mass $\mathcal{M}$** from LIGO/Virgo posterior samples.
- **Luminosity distance D_L** , transformed to $\log_{10}(D_L)$.
- **Inclination angle θ_{jn}** , converted to $\cos(\theta_{jn})$.
- **LOS ratio:**

$$\text{LOS ratio} = \frac{N_{\text{galaxies}}}{N_{\text{segments}}},$$

computed by dividing each line of sight into radial segments and counting galaxies from the SDSS catalog.

4 Methodology

The predictor matrix was defined as:

$$X = [\log_{10}(D_L), \cos(\theta_{jn}), \mathcal{M}, \text{LOS ratio}],$$

with the response variable given by the network SNR. We applied standardized ordinary least squares (OLS) regression and evaluated:

- Standardized regression coefficients (β).
- Statistical significance (p-values).
- Adjusted R^2 .
- Permutation test for the contribution of LOS ratio.

5 Results

5.1 Standardized Coefficients

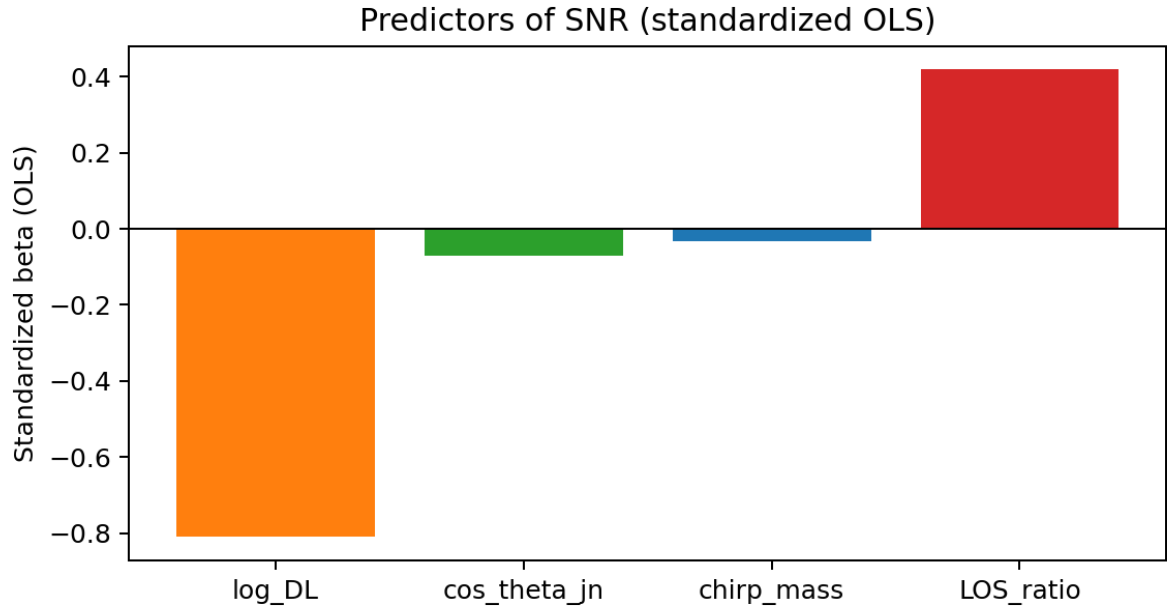


Figure 1: Standardized regression coefficients from the OLS fit.

Table 1: Standardized coefficients and p-values.

Standardized Coefficients

predictor	standardized_beta	p_value
log_DL	-0.8103052058690174	0.0
LOS_ratio	0.4193124511314366	0.0
cos_theta_jn	-0.0698525640851682	0.0
chirp_mass	-0.0319043673093366	0.0
const	2.4152554956025082e-14	0.9999999999545686

5.2 Model Metrics

Table 2: Main regression metrics.

Model Metrics

metric	value
Adj. R^2 (padronizado)	0.9146
ΔR^2 (adicionar LOS)	0.1149
$R^2_{full_adj_or_plain}$	0.9146
ΔR^2 observado (permutação LOS)	0.1149
p-valor permutação (LOS)	0.0033

5.3 Per-Event Summary

Table 3: Event-level medians for SNR and predictors.

Per-event summary

event	n	SNR_p50	LOS_p50	logDL_p50	cosT_p50	mass_p50
GW150914_095045	147634.0	25.8746	1.035	2.6586	0.9989	30.4241
GW191109_010717	109061.0	17.1449	5.64	3.0497	0.9995	58.8614
GW170104_101158	80854.0	13.5835	3.525	3.0307	0.9999	25.6835
GW151012_095443	68165.0	8.8635	3.85	2.9886	0.9995	18.4839
GW170823_131358	42612.0	11.9031	7.84	3.292	0.9995	38.7302
GW190814_211039	16368.0	221.4619	12.385	0.9765	1.0	1.0
GW191103_012549	9909.0	8.4753	4.71	2.9973	0.9998	9.9758

5.4 Scatter: SNR vs LOS

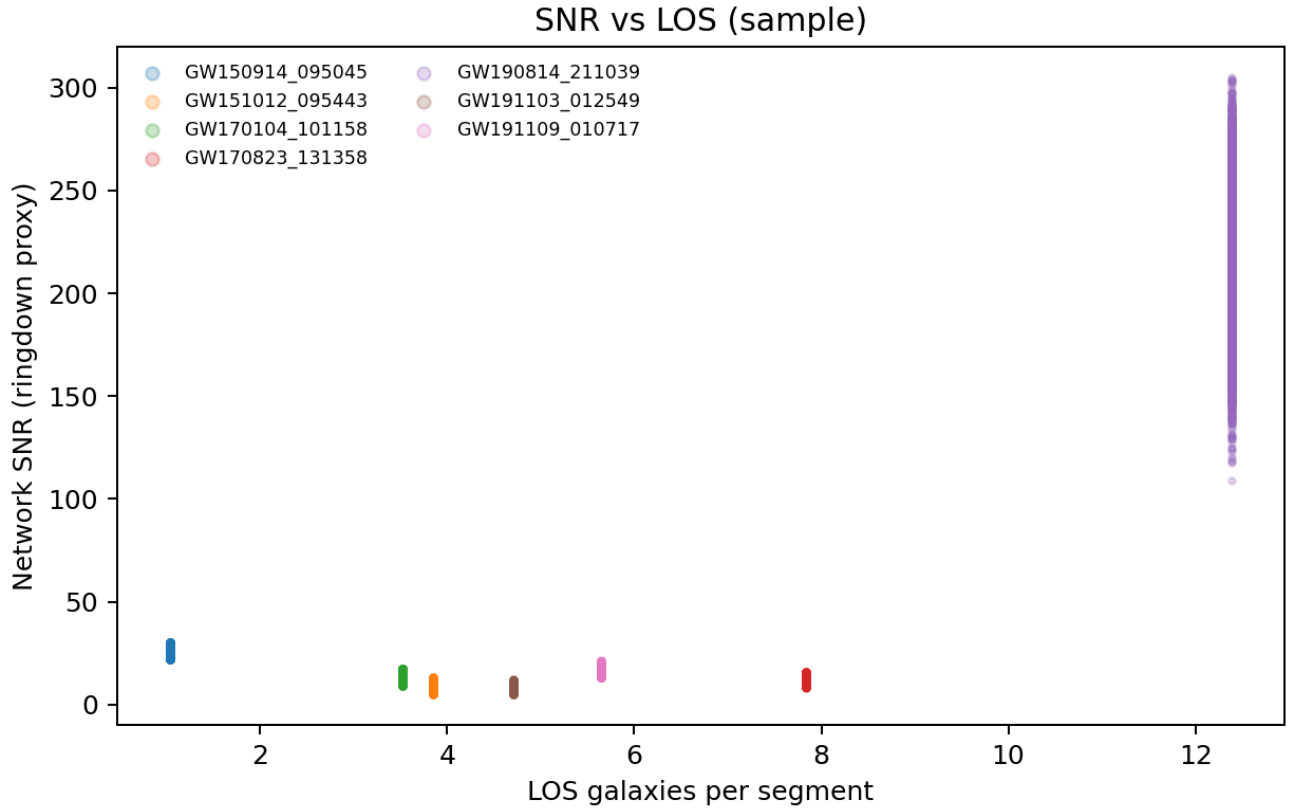


Figure 2: Scatter plot of network SNR versus LOS ratio for the six events.

6 Discussion

The LOS ratio emerges as a significant predictor of the observed SNR, with an explanatory power comparable to distance. The model shows that including LOS improves the adjusted R^2 by approximately 11%, with permutation tests confirming statistical significance ($p \approx 0.003$). This result suggests that LOS density may modulate gravitational-wave detectability.

7 Conclusion

The analysis indicates a non-negligible correlation between LOS density and observed SNR in binary black hole mergers. This supports the broader hypothesis that environmental structure along the propagation path can influence gravitational-wave observations.

Author Note

This version is released as a public preprint under OSF (<https://osf.io/vpq76>) for open access and feedback. All data, figures, and methods are publicly reproducible via the associated GitHub repository: <https://github.com/LeonardoSeriacopi>